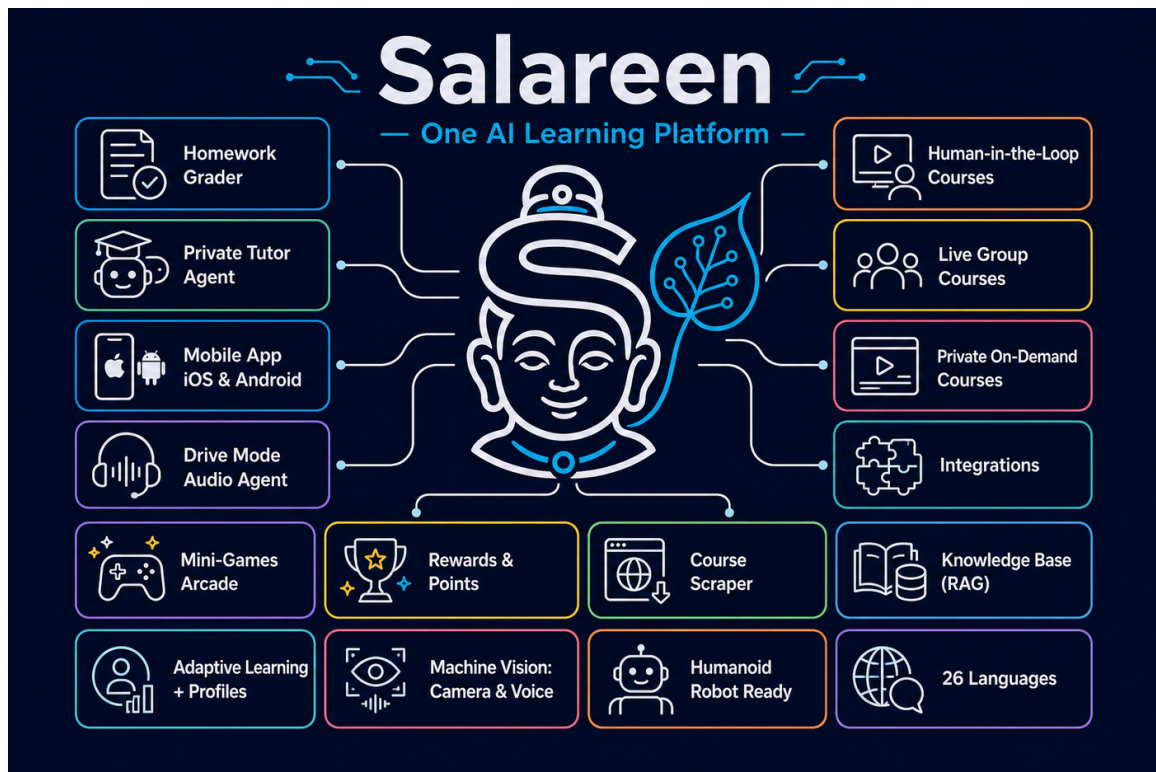


Salareen

Agentic Online Education Platform

Platform Architecture, Features & Capabilities Overview



Platform ecosystem map

Version 0.43.0 · Generated 2026-07-24

Compiled from the live monorepo: FastAPI microservices, Next.js web, Expo mobile, aoep_shared providers, billing/payments, and product screens under docs/screens/.

1. Executive summary

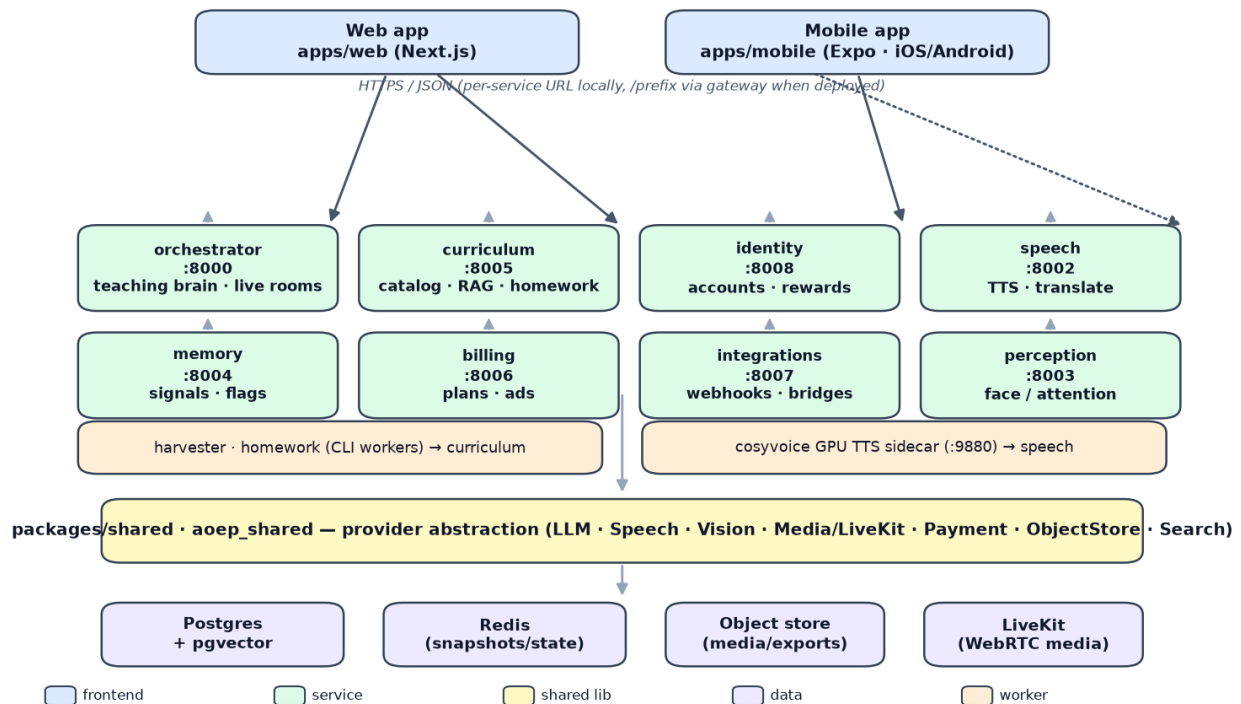
Salareen (also referred to as AOEP / AI Classroom in parts of the codebase) is a multi-service education platform for live AI-taught classes, Drive Mode audio lessons, adaptive learning, language learning, corporate upskilling, arcade games, rewards, careers matching, compliance controls, and third-party integrations. The same codebase runs **local**, **cloud**, or **edge** by environment configuration only — no code forks.

- **Version:** 0.43.0 (repo VERSION)
- **Clients:** Next.js web (apps/web) + Expo mobile (apps/mobile)
- **Backend:** FastAPI microservices behind aoep_shared providers
- **Realtime media:** LiveKit (Salareen group/solo rooms)
- **Deploy:** Vultr Kubernetes Engine (VKE) + Cloudflare DNS/TLS
- **Monetization:** Free / Standard (\$19.99) / VIP Premium (\$29.99) + ads + global payments

2. Logical architecture

Browser and mobile apps call HTTPS JSON APIs. Locally each service has its own port; in cluster every service listens on :8000 and is reached via path prefixes (/identity, /curriculum, ...). Shared business logic lives in packages/shared (distribution aoep-shared).

1 · System topology — frontend, backend, shared, data



System topology (services, clients, data, providers)

3. Services & ports

Service	Pkg	Port	Owns
orchestrator	orchestrator	8000	Teaching brain: lessons, sessions, assessment, live rooms, group classes, director
speech	speech_gw	8002	TTS routing, translation, language learning APIs
perception	perception	8003	Consent-gated face recognition / engagement (YuNet+SFace)
memory	memory	8004	Profiles, consent, legal, feature flags, surveys, telemetry
curriculum	curriculum	8005	Catalog, RAG, homework, scenes, ads plan, recommendations
billing	billing	8006	Plans, entitlements, checkout, ad slots / revenue ledger
integrations	integrations	8007	Webhooks, LMS/LTI, finance, Zoom/Teams/Meet bridges
identity	identity	8008	Accounts, auth, students, rewards, enrollments, language
harvester	harvester	CLI	Crawl/generate courses; mandatory .pptx + .course.json export
homework	homework	CLI	Generate / OCR / authorship / autograde worker
cosyvoice	sidecar	9880	GPU CosyVoice 2 TTS for clone narration

Every HTTP service exposes `/health`, `/version`, `/__meta`, `/metrics`, and `/telemetry/*` via `create_service()`.

4. Product feature catalog

4.1 Learning surfaces

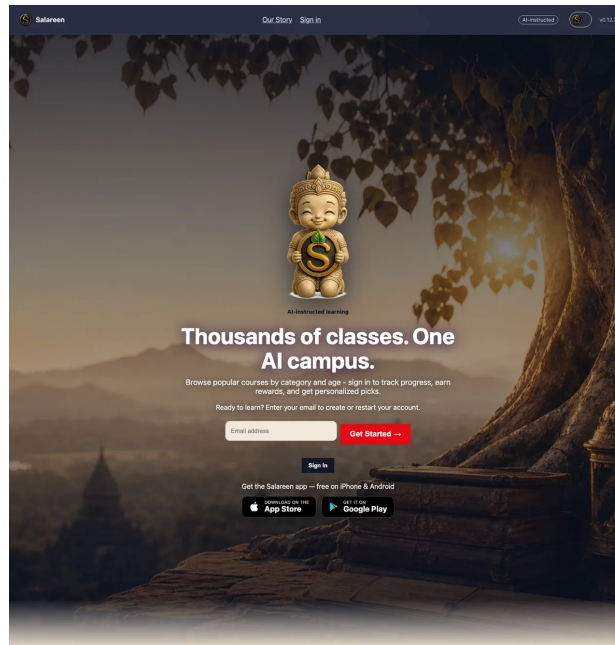
- **Live AI class** — Theodore narrates slides; Ask-the-teacher with RAG-grounded answers
- **Group classes / live rooms** — LiveKit video, raise-hand Q&A, gifts, moderation
- **Solo 1:1 rooms** — same room model scaled to one learner
- **Drive Mode** — eyes-free audio courses with hands-free voice Q&A
- **Corporate / professional courses** — mid-course pop quiz + required end assessment
- **Homework grader** — typed or handwritten with rationale + citations
- **Human-in-the-loop** — educator console reviews / co-grades when confidence is low
- **Languages** — pronunciation, speech practice; UI in 27 languages
- **Kids Mode** — age-appropriate pathways and parental account model
- **Arcade** — quiz/match + canvas games (geometry, stocks, chemistry Potion Lab, Challenge the AI)
- **Careers / jobs** — JD parse, skills match, certifications
- **XR lab** — audience readiness / XR assess APIs
- **Vision** — opt-in face/engagement via perception service

4.2 Platform & ops

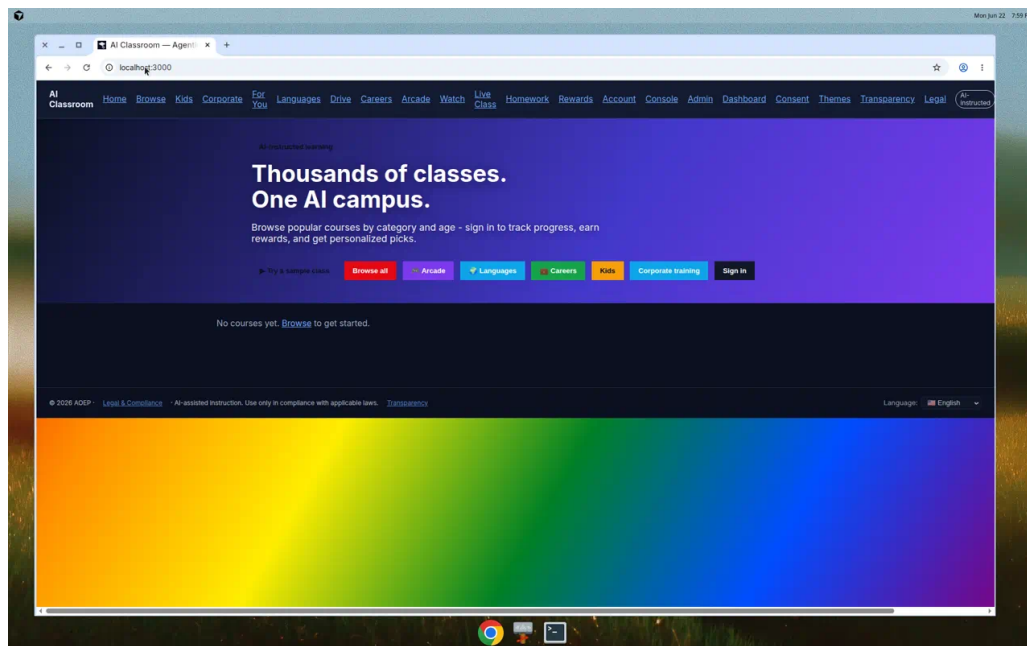
- Netflix-style home / browse / watch / recommended feeds
- Membership tiers, billing checkout, entitlements gate
- Rewards & points ledger with prize redemption
- Feature flags (admin) with per-subject overrides
- Post-class surveys + admin insights / data mart
- Consent, legal, transparency, model cards, security pages
- Admin observability (versions, telemetry, ad revenue)
- Content packs + harvester corpus growth without code forks
- Adaptive learning / Foresight recommendations
- Assessment: formative checkpoints + verified summative pass tokens

5. Product screenshots

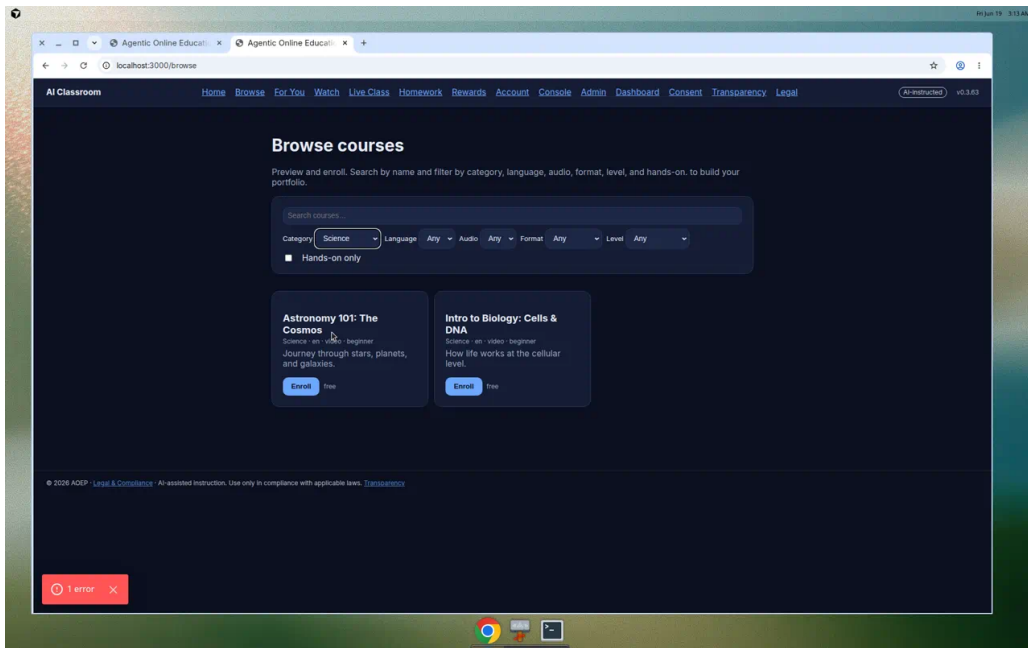
Screens captured from the shipped web UI (docs/screens/). Newer arcade canvas games may not yet have dedicated screenshots; quiz/match and core learning surfaces are shown.



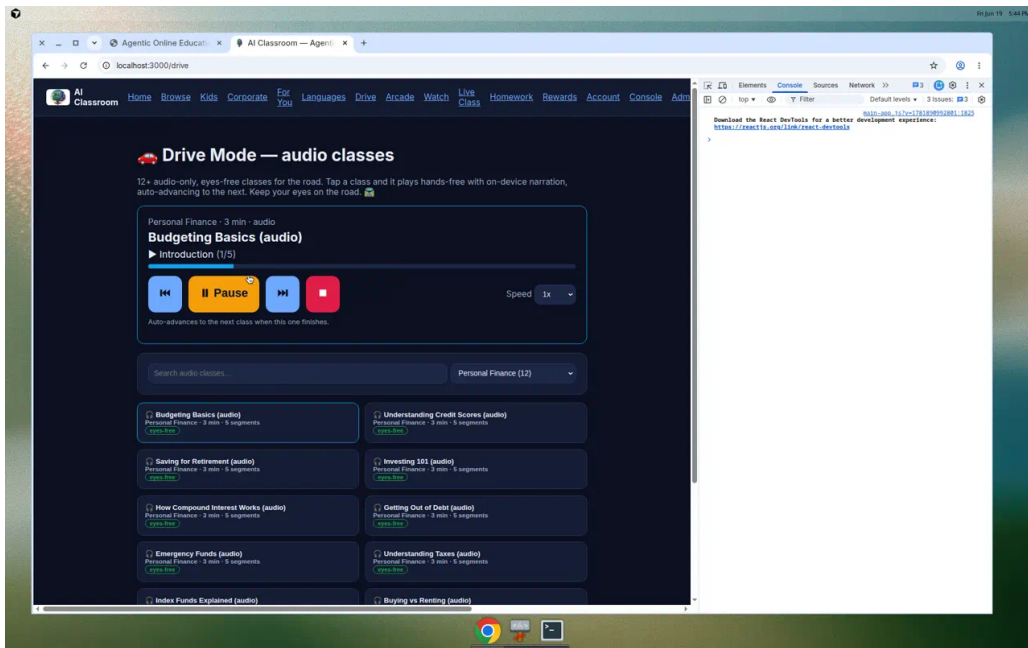
Marketing / landing



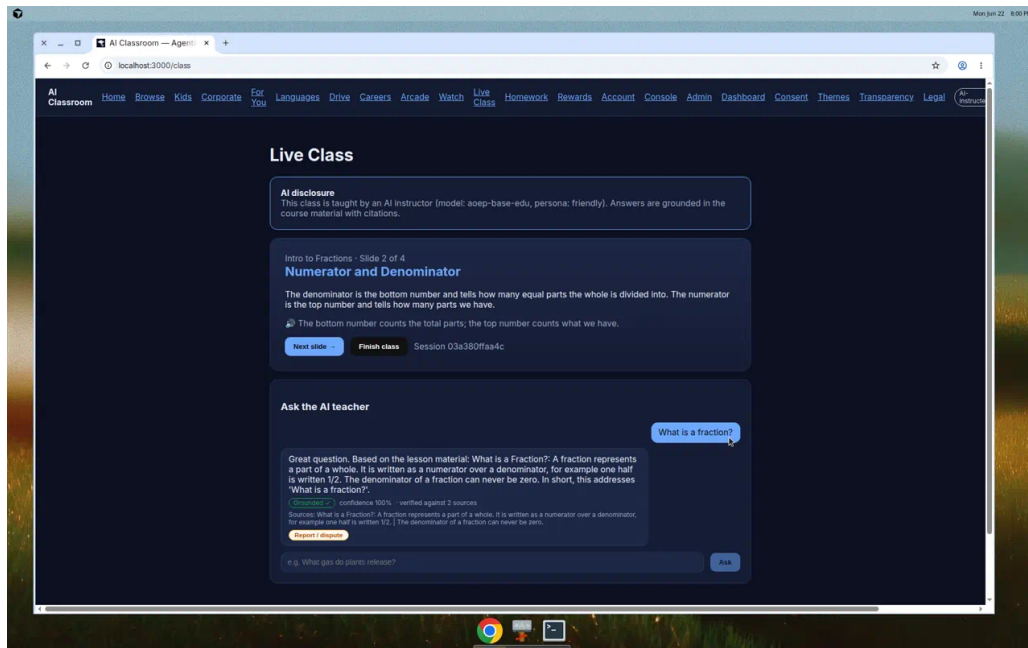
Signed-in home feed



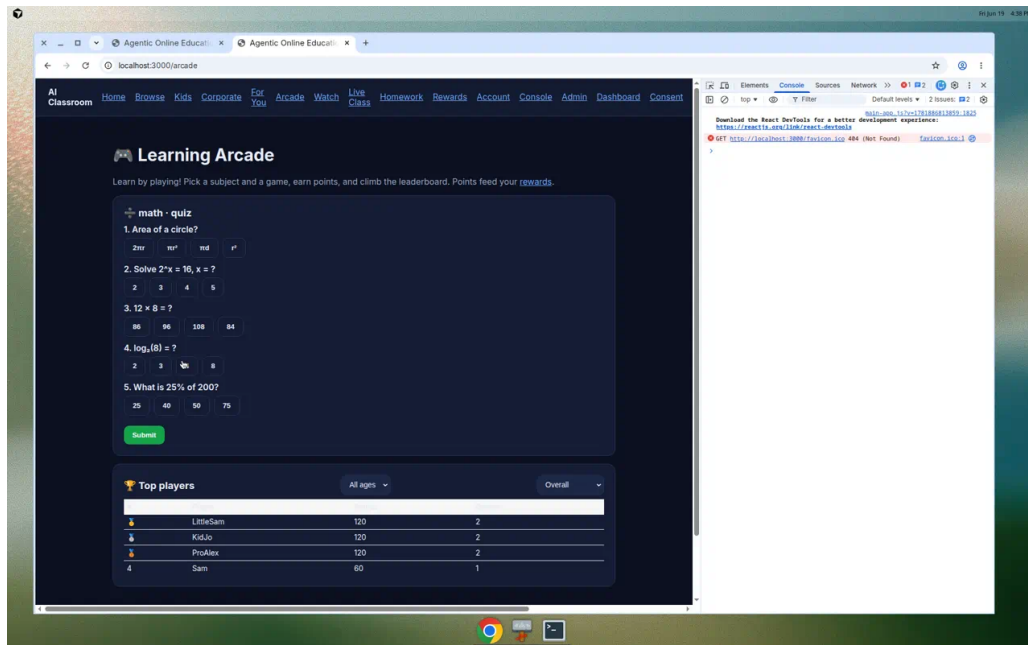
Browse catalog



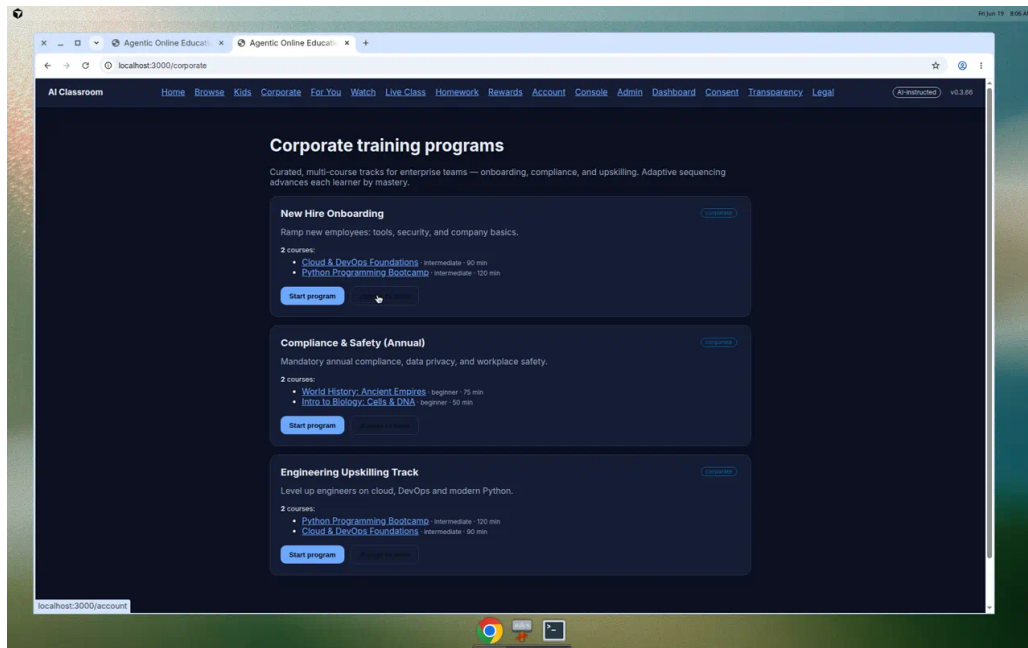
Drive Mode player



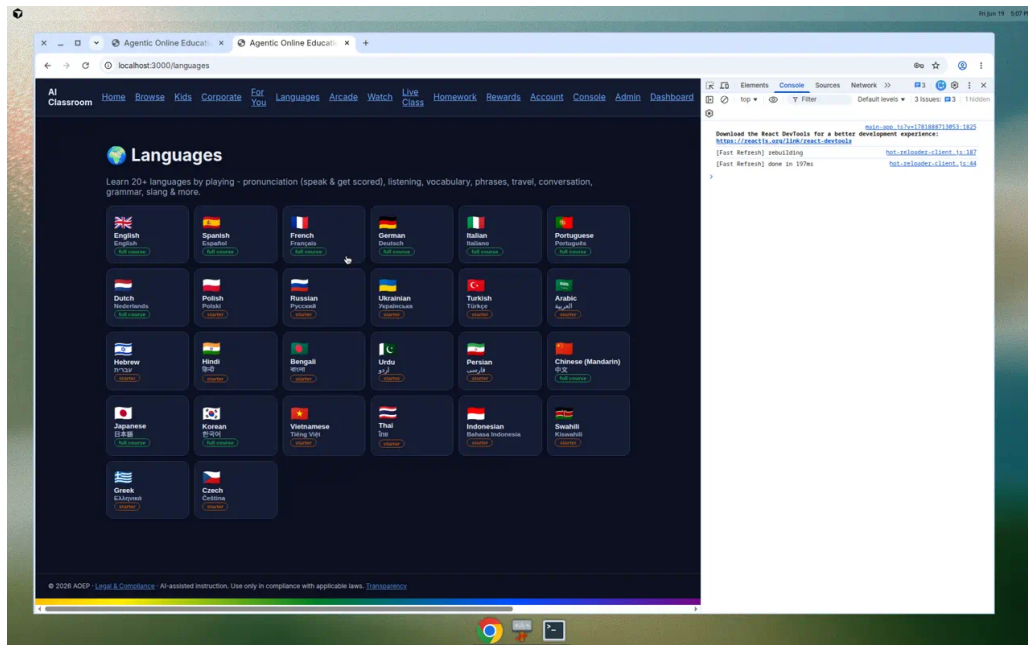
Live class + grounded Ask answer



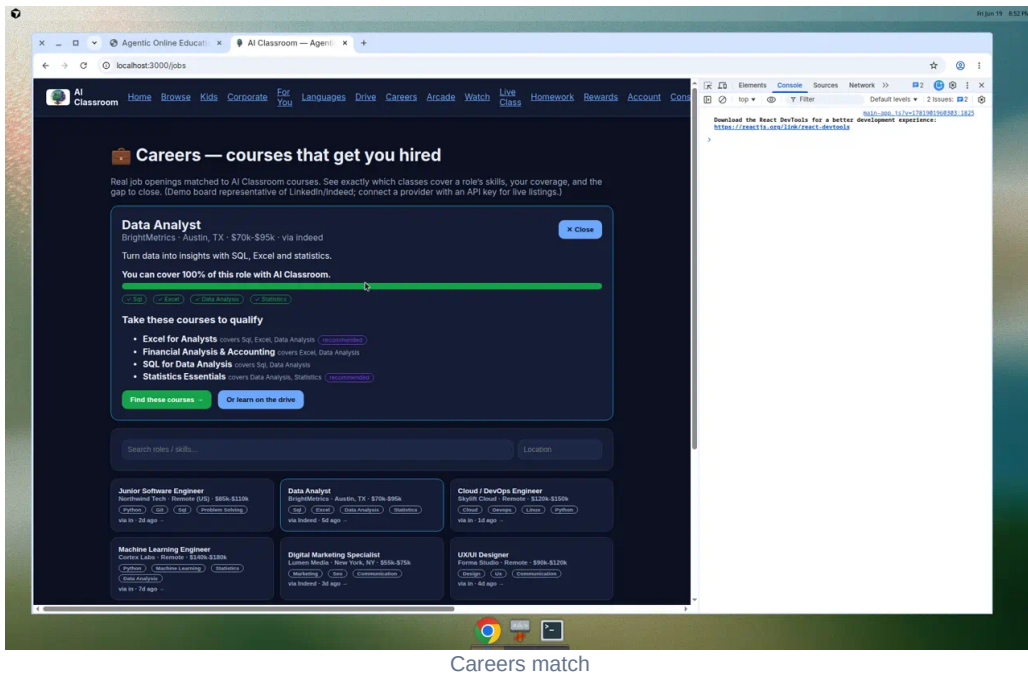
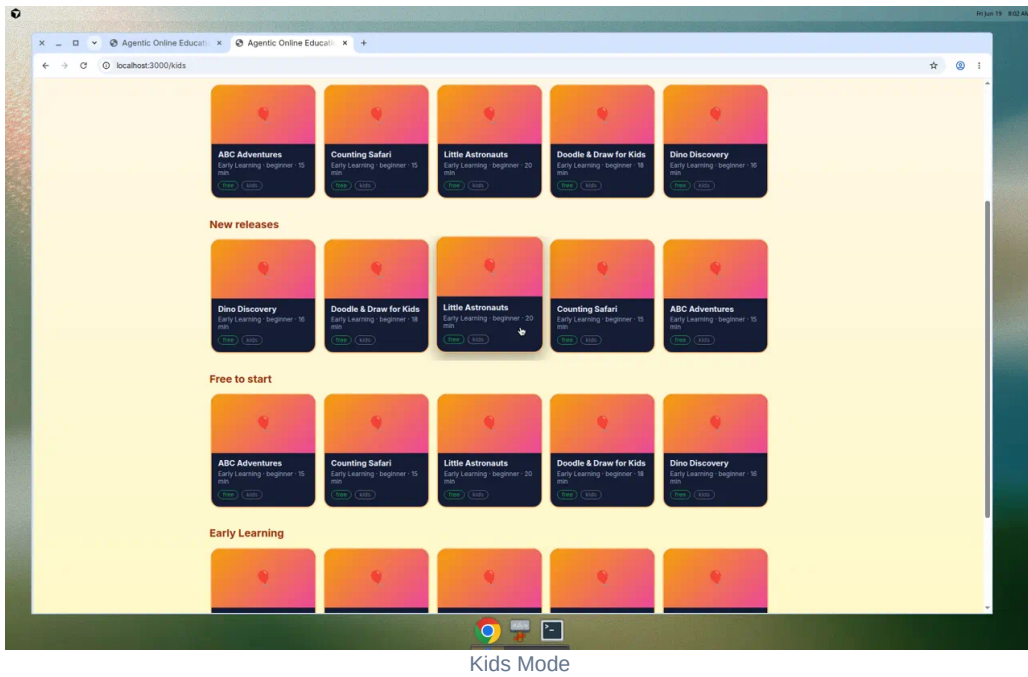
Learning Arcade quiz

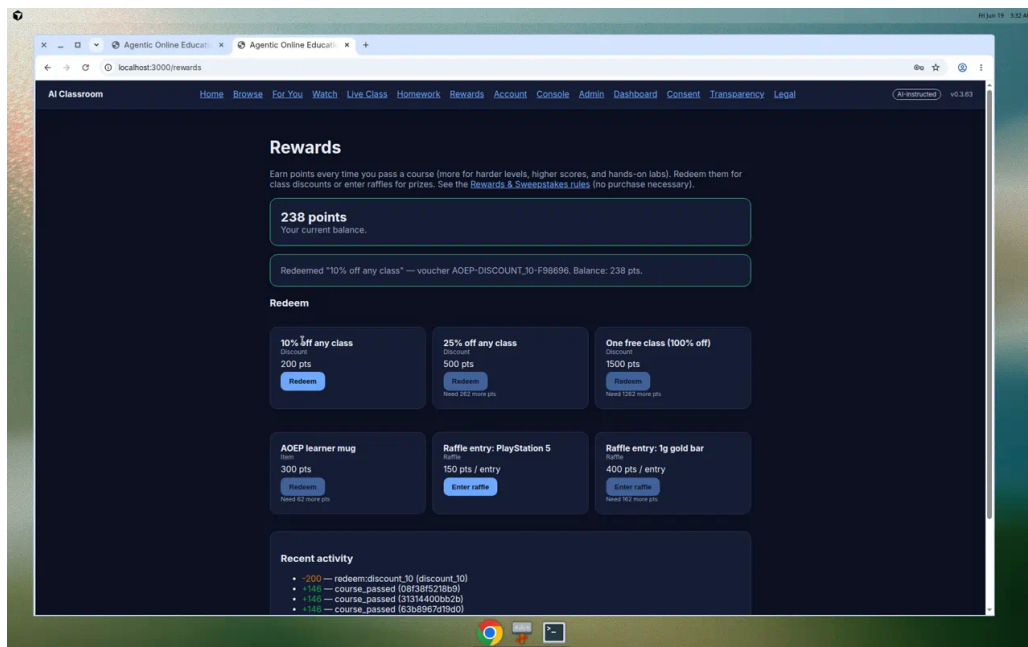


Corporate training programs

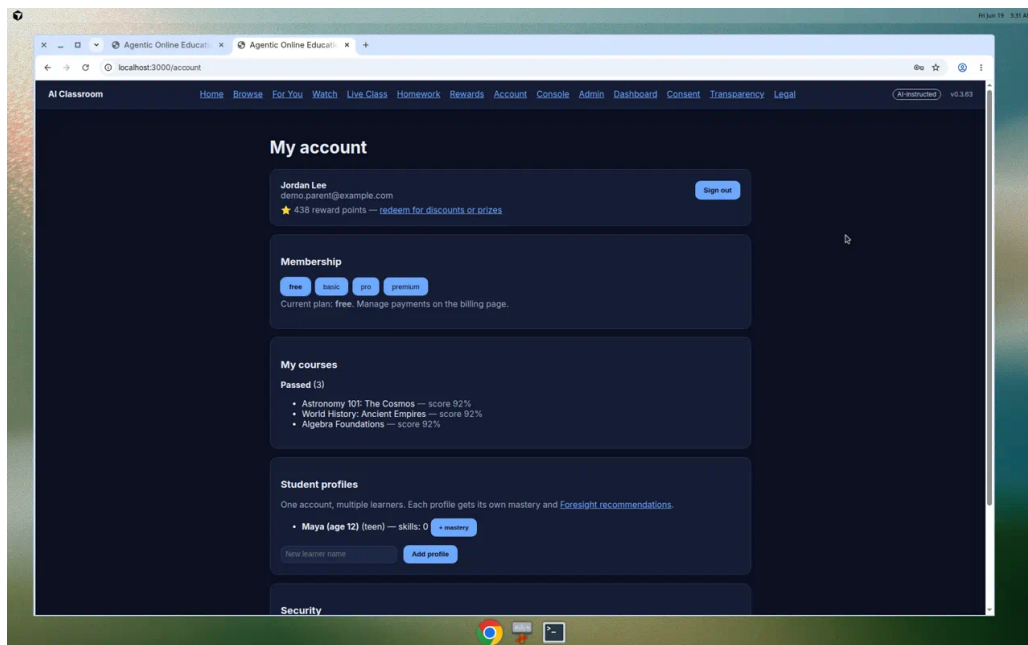


Languages grid

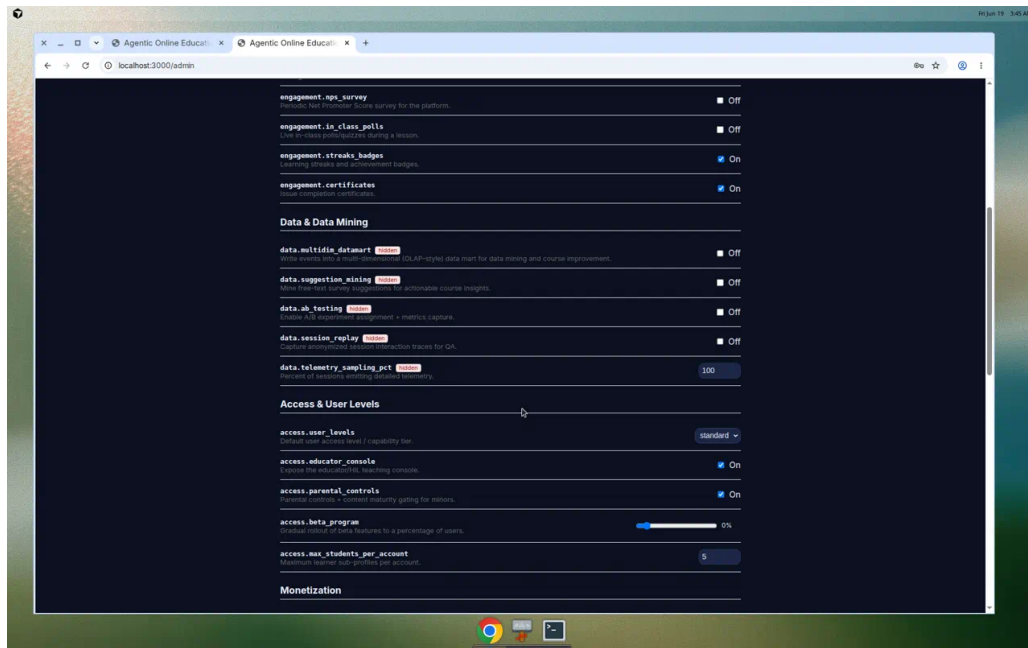




Rewards redeem



Member account



Admin feature flags

6. Provider capabilities (dual-mode)

Capabilities are selected by DEPLOY_MODE (local | cloud | edge) plus optional per-component <COMPONENT>_MODE. Blank override inherits deploy mode. Offline paths degrade gracefully (RAG tutor fallback, device TTS, sandbox payments).

Capability	Local / offline	Cloud
LLM	vLLM/Ollama or grounded RAG fallback	Cloud LLM endpoint
Speech / TTS	edge-tts → device voices; CosyVoice optional	ElevenLabs → edge-tts → device
Vision	OpenCV YuNet + SFace (CPU)	Same self-hosted models
Media	Local LiveKit or degraded tiles	LiveKit Cloud / cluster JWT mint
Object store	Local / MinIO	S3-compatible (Vultr Object Storage)
Payment	Sandbox (all methods simulated)	Routed real processors (keys required)
OCR / Search / Jobs	Local stubs / light impls	Cloud connectors when configured

6.1 Feature flags

Runtime catalog currently defines **52** flags across engagement, data, access, monetization, AI, UX, and ops — evaluated via memory service (GET /flags/evaluate) with admin overrides.

7. Membership & payment systems

7.1 Consumer plans

Netflix-style calendar billing on the signup day. Standard includes ads; VIP Premium is ad-free. Entitlements are gated by the billing service (`can_start`) before classes start.

Plan	Tier	Price (USD/mo)	Notes
Free	free	\$0.00	English classes with ads
Standard	basic	\$19.99	5 languages, billed monthly on your signup day
VIP	premium	\$29.99	All languages, ad-free, analytics — billed monthly on your signup day

7.2 Global payment methods

The platform models **50** consumer payment methods routed across **13** processors. Local sandbox simulates every method; cloud providers advertise only what their API keys can process. Country/locale helpers pick a sensible default method set for checkout UI.

Processor	Example methods
aba	aba_pay, khqr, wing
local_psp	fawry, knet, line_pay, mada, payeasy, stc_pay, unionpay
manual	zelle
mercado_pago	mercado_pago, pix
momo	momo, zalo_pay
paypal	paypal, venmo
paytm	paytm
razorpay	phonepe, razorpay, rupay, upi
square	square
stripe	ach, affirm, afterpay, alipay, apple_pay, bancontact, boleto, card, cashapp, eps, giropay, google_pay...
toss	kakao_pay, naver_pay, toss
vnpay	vnpay
yoomoney	mir, yoomoney

Regional coverage includes Stripe rails (cards, wallets, BNPL, SEPA, Alipay/WeChat, ...), PayPal/Venmo, Square, Razorpay (UPI/PhonePe), Mercado Pago (PIX), VNPAY/MoMo, ABA/KHQR/Wing (Cambodia), YooMoney/Mir, Toss/Kakao/Naver, and manual Zelle.

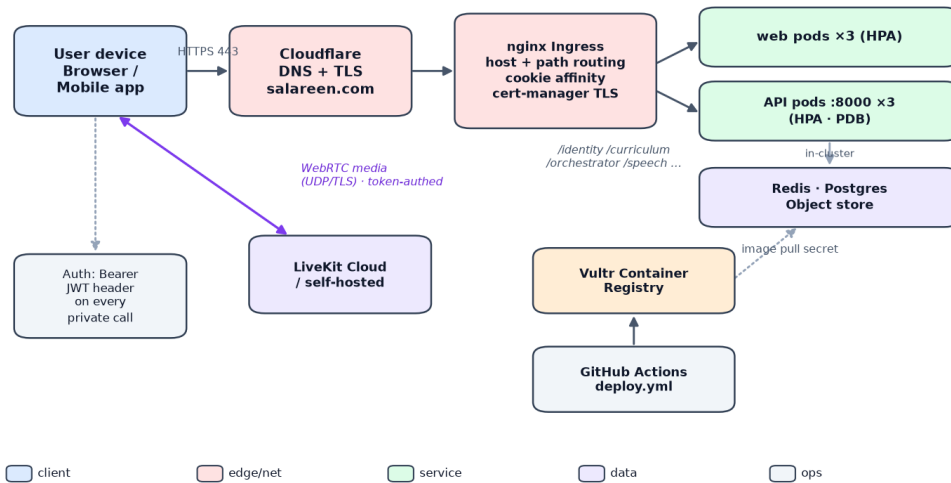
7.3 Ads & revenue

- Video pre/mid-roll (VAST/VMAP) on watch / class / Drive — empty plan for VIP
- Display slots via billing /ads/slot; mobile AdMob banners
- Impression/click beacons → AdRevenueLedger; admin Ad Revenue panel

8. Key system flows

Network traffic

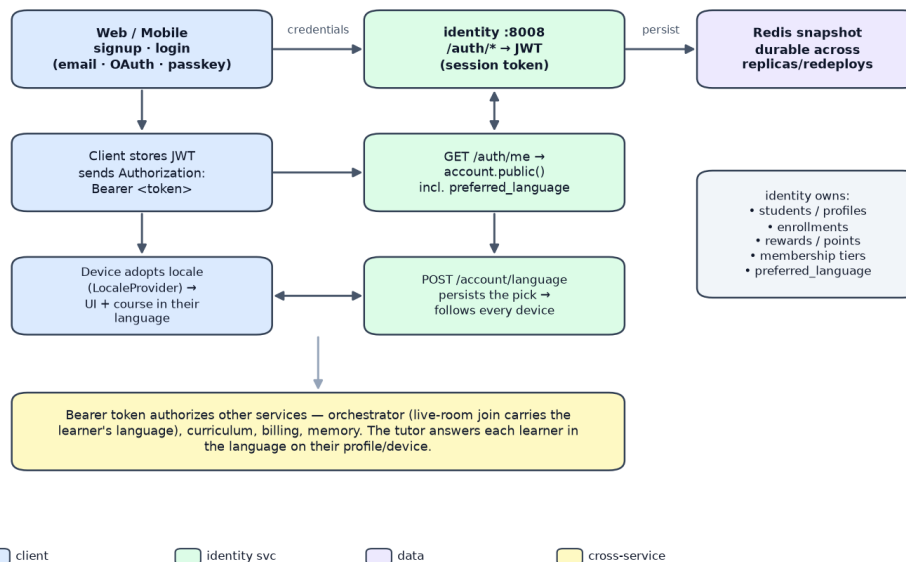
2 · Network traffic flow — device → edge → services → data



Cloudflare → nginx Ingress (path routing + session affinity) → web/API pods; WebRTC to LiveKit is a separate path.

Accounts & language

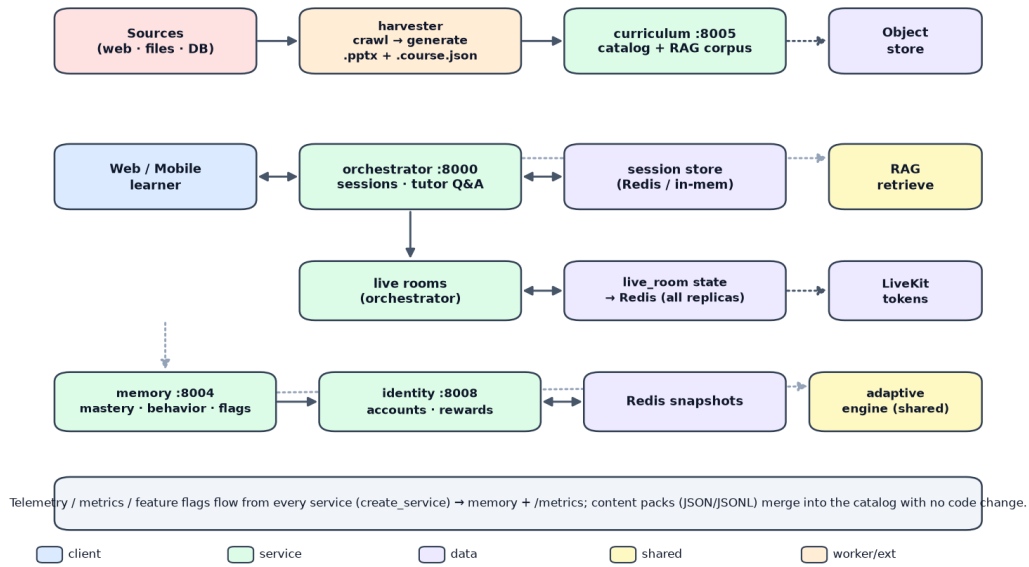
3 · Accounts, auth & language — identity of the learner



Identity mints JWT; preferred_language follows the learner across devices; rewards and membership live here with Redis snapshot durability.

Data & content

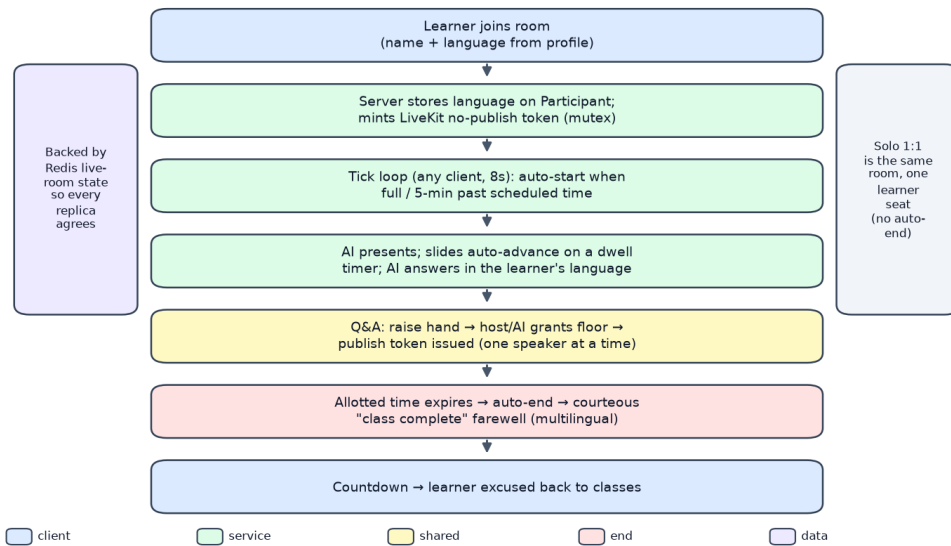
4 · Data flow & stores — content, sessions, signals, state



Harvester → curriculum catalog/RAG; orchestrator sessions; Redis room state; object storage for media/exports.

Live-class lifecycle

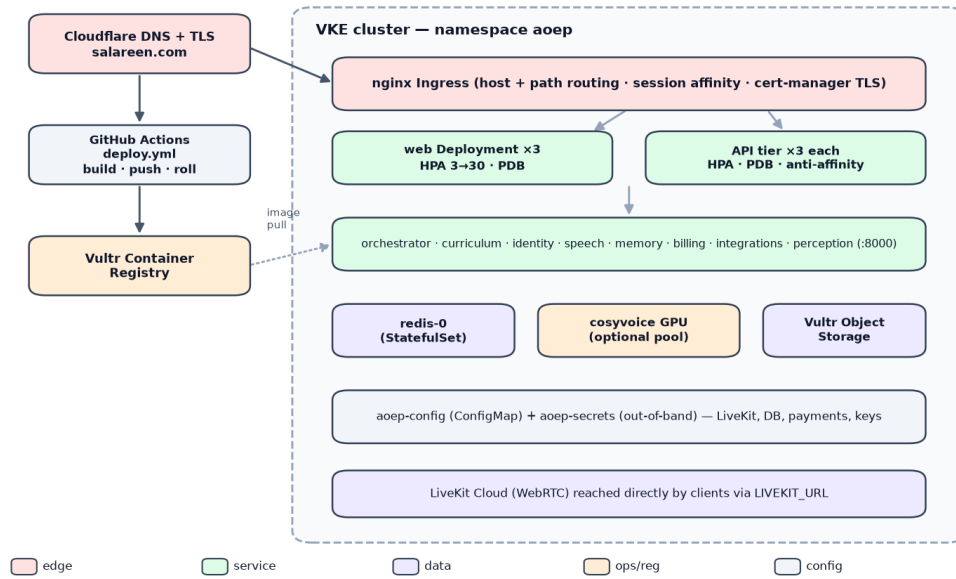
5 · Live-class lifecycle — join → teach → Q&A → complete



Join → LiveKit token → tick loop auto-start/advance/end; raise-hand speaking mutex.

VKE deployment

6 · Deployment — Vultr Kubernetes Engine (namespace aoep)



GitHub Actions builds to Vultr registry; HPA/PDB; ConfigMap + out-of-band secrets.

9. Client surfaces

9.1 Web (Next.js)

73 top-level app routes (dynamic segments omitted), including arcade suite, corporate learn, Drive Mode, live rooms, billing, admin, and legal.

```
/, /account, /admin, /admin/presence, /arcade, /arcade/ai-duel, /arcade/card-match, /arcade/challenge-ai, /arcade/challenge-ai/connect-four, /arcade/challenge-ai/grid-master, /arcade/challenge-ai/number-duel, /arcade/challenge-ai/quiz-duel, /arcade/challenge-ai/tic-tac-toe, /arcade/chemistry, /arcade/cosmic-catch, /arcade/creature-catch, /arcade/geo-blocks, /arcade/geometry-blocks, /arcade/geometry-tetris, /arcade/hidden-items, /arcade/jeopardy, /arcade/kart-race, /arcade/market-catch, /arcade/market-moves, /arcade/photo-reveal, /arcade/pro-scenarios, /arcade/reef-quest, /arcade/shape-drop, /arcade/shape-stack, /arcade/solar-3d, /arcade/species-match, /arcade/spot-difference, /arcade/stem-research, /arcade/stock-rush, /arcade/stock-trader, /arcade/stocks, /arcade/uno-quiz, /arcade/zoo-safari, /backgrounds, /billing, /browse, /class, /consent, /console, /contact...
```

9.2 Mobile (Expo)

27 screens: AccountScreen, ArcadeScreen, AudioCoursesScreen, AuthScreen, BillingScreen, BugReportScreen, CareersScreen, CorporateScreen, DemoScreen, DriveModeScreen, GameScreen, GroupClassesScreen, HomeScreen, KidsScreen, LanguagesScreen, LessonScreen, LiveClassScreen, LiveRoomScreen, LiveRoomsScreen, MyListScreen, NotificationsScreen, PotionLab, RewardsScreen, SearchScreen, SecurityScreen, SettingsScreen, WorldsScreen.

10. Security, privacy & compliance posture

- Auth: PBKDF2 credentials + HMAC/JWT bearer tokens (identity)
- Admin ops gated by ADMIN_SECRET
- Biometrics opt-in + consent-gated; self-hosted vision models
- Legal acceptance, retention, transparency, and model-card pages
- Redis snapshot durability for identity (not a throwaway cache)
- Secrets managed out-of-band in k8s; ConfigMap holds non-secret config

11. How to regenerate

```
pip install matplotlib==3.11.0 reportlab==5.0.0 Pillow==11.1.0
python3 scripts/build_architecture_pdf.py — diagrams + docs/architecture.pdf
python3 scripts/build_platform_overview_pdf.py — this overview PDF
```

Canonical text: docs/architecture.txt. API surface: docs/api-reference.txt. Hosting: docs/hosting.txt /
infra/k8s-vke/RUNBOOK.txt.